NST-07

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NEET SPECIAL GRAND TEST - 07 (09-04-2026)

(P – 1 & P – 2) SR ELITE, AIIMS S60, MPL & LTC VAIDYAH

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on side-1 and side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction, electronic/manual calculator are NOT permissible.
8. Each candidate must show on demand his/her Admit Card to the Invigilator.
9. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over there Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
13. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

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PHYSICS

1. Two identical charged spheres of material density ρ, suspended from the same point by inextensible strings of equal length make an angle θ between the strings. When suspended in a liquid of density σ the angle θ remains the same. The dielectric constant K of the liquid is

(1) $\frac{\rho}{\rho-\sigma}$ (2) $\frac{\rho-\sigma}{\rho}$

(3) $\frac{\rho}{\rho+\sigma}$ (4) $\frac{\rho+\sigma}{\rho}$

2. Four electric point charges $+q,+q,-q$ and $-q$ are placed at the corners of a square of side 2L. The electric potential at point A, midway between the two charges +q and +q, is

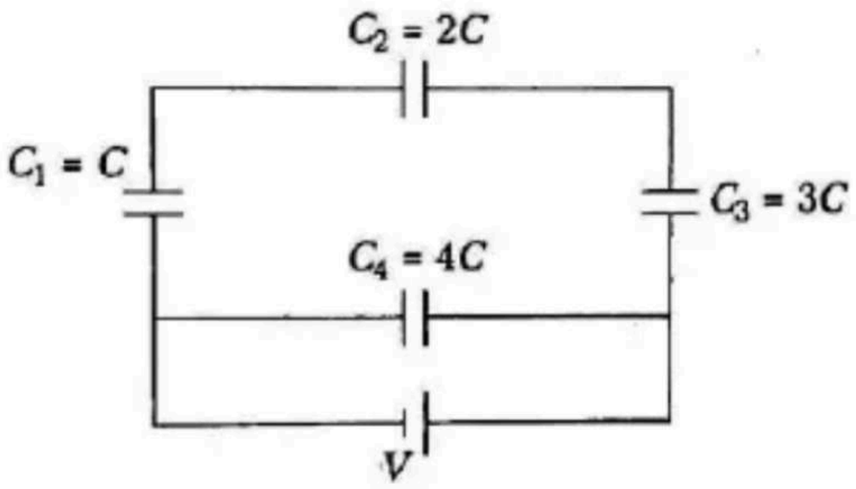
(1) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1+\frac{1}{\sqrt{5}}\right)$

(2) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1-\frac{1}{\sqrt{5}}\right)$

(3) Zero

(4) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1+\sqrt{5}\right)$

3. A network of four capacitors of capacity equal to $C_1=C, C_2=2C, C_3=3C$ and $C_4=4C$ are connected to in a battery as shown in the figure. The ratio of the charges on C_2 and C_4 is



(1) $\frac{22}{3}$ (2) $\frac{3}{22}$

(3) $\frac{7}{4}$ (4) $\frac{4}{7}$

4. Three resistances P, Q, R each of 2Ω and an unknown resistance S form the four arms of a Wheatstone's bridge circuit. When a resistance of 6Ω is connected in parallel to S, the bridge gets balanced. What is the value of S?

(1) 2Ω (2) 3Ω

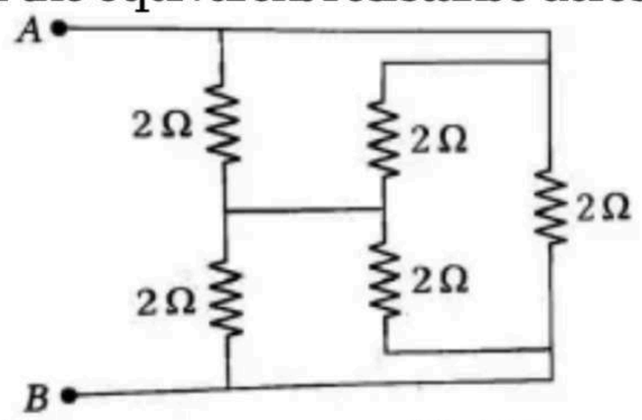
(3) 6Ω (4) 1Ω

5. A cell supplies a current i_1 through a resistance R_1 and a current i_2 through a resistance R_2. The internal resistance of this cell is

(1) R_2-R_1 (2) $\frac{i_1R_2-i_2R_1}{i_1-i_2}$

(3) $\frac{i_2R_2-i_1R_1}{i_1-i_2}$ (4) $\left(\frac{i_1+i_2}{i_1-i_2}\right)\sqrt{R_1R_2}$

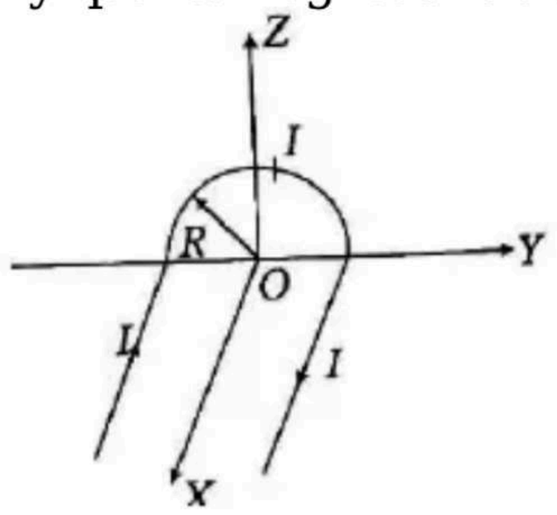
6. Find the equivalent resistance across AB.



(1) 1Ω (2) 2Ω

(3) 3Ω (4) 4Ω

7. A wire carrying current I has the shape as shown in adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semi-circular portion of radius R is lying in yz-plane. Magnetic field at point O is



(1) $B=\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}+2\hat{k}\right)$

(2) $B=-\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}-2\hat{k}\right)$

(3) $B=-\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}+2\hat{k}\right)$

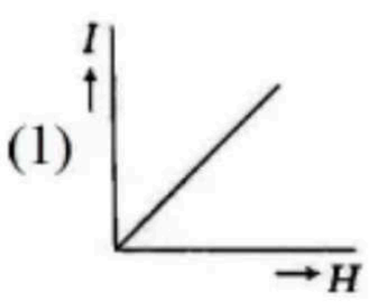
(4) $B=\frac{\mu_0}{4\pi}\frac{I}{R}\left(\pi\hat{i}-2\hat{k}\right)$

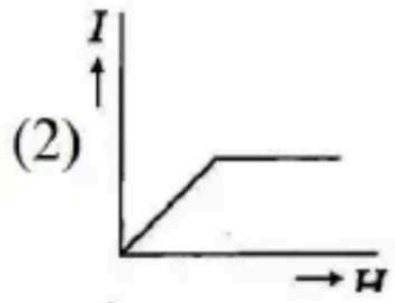
8. A proton is moving in a uniform magnetic field B in a circular path of radius a in a direction perpendicular to Z-axis along which field B exists. Calculate the angular momentum. If the radius is a and charge on proton is e.

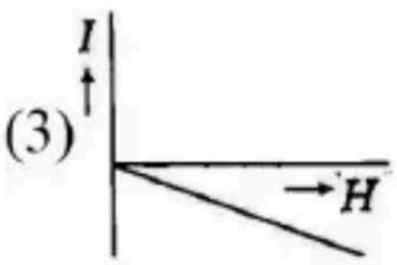
(1) $\frac{Be}{a^2}$ (2) eB^2a

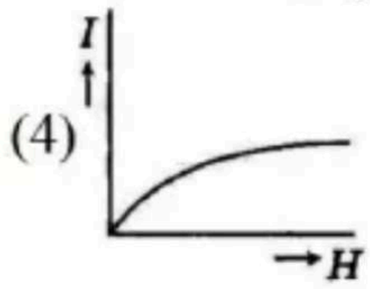
(3) a^2eB (4) aeB

9. The correct curve between intensity of magnetisation (I) and magnetic field (H) for a paramagnetic substance is given by

(1) 

(2) 

(3) 

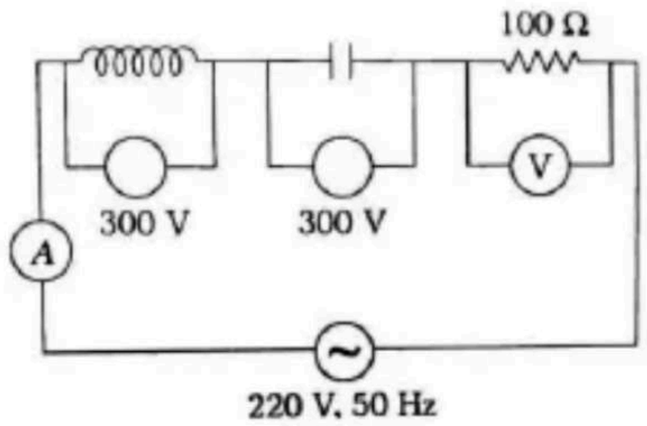
(4) 

10. The inductance of a coil in which a current of 0.1 A increases at the rate of 0.5 A/s and the power flow is 0.5 watt, is

(1) 2 H (2) 4 H

(3) 8 H (4) 10 H

11. In the circuit shown below what will be the reading of the voltmeter and ammeter?



(1) 200 V, 1 A (2) 800 V, 2 A

(3) 220 V, 2 A (4) 220 V, 2.2 A

12. The induced emf can be produced in a coil by

(A) Moving the coil with uniform speed inside magnetic field

(B) Moving the coil with non uniform speed inside uniform magnetic field

(C) Rotating the coil inside the uniform magnetic field

(D) Changing the area of the coil inside the uniform magnetic field

Choose the correct answer from the options given below.

(1) B and D only (2) B and C only

(3) A and C only (4) C and D only

13. The frequencies of X-rays, $\gamma - rays$ and ultraviolet rays are respectively p, q and r, then

(1) $p > q, q > r$ (2) $p < q, q > r$

(3) $p < q, q < r$ (4) $p > q, q < r$

14. Two plane mirrors are inclined at 60° to each other. If an object is placed in between them, then the number of images will be

(1) 5 (2) 8

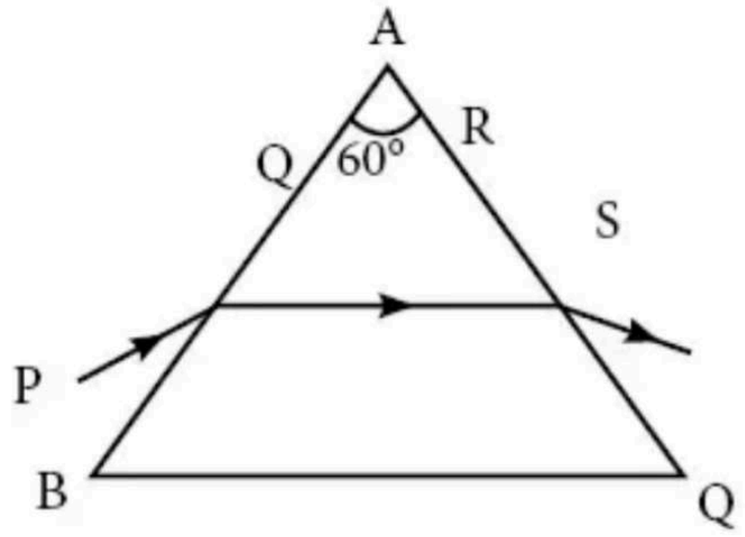
(3) 6 (4) 9

15. An object is placed 30 cm away from a convex lens of focal length 10 cm and a sharp image is formed on a screen. Now, a concave lens is placed in contact with the convex lens. The screen now has to be moved by 45 cm to get a sharp image again. The magnitude of focal length of the concave lens is (in cm)

(1) 72 (2) 60

(3) 36 (4) 20

16. A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS, such that AQ = AR. If the angle of prism A $= 60^0$ and the refractive index of the material of prism is $\sqrt{3}$, then the angle of deviation of the ray is



(1) 60^0 (2) 45^0

(3) 30^0 (4) 90^0

17. The intensity of each of the two slits in Young's double slit experiment is I_0. Calculate the minimum separation between the two points on the screen where intensities are $2I_0$ and I_0. Given the fringe width equal to α.

(1) $\frac{\alpha}{4}$ (2) $\frac{\alpha}{3}$

(3) $\frac{\alpha}{12}$ (4) $\frac{\alpha}{8}$

18. Find the correct statement(s) about photoelectric effect.

(a) There is no significant time delay between the absorption of a suitable radiation and the emission of electrons.

(b) Einstein analysis gives a threshold frequency below which no electron can be emitted.

(c) The maximum kinetic energy of the emitted photoelectrons is proportional to the frequency of incident radiation.

(d) The maximum kinetic energy of electrons does not depend on the intensity of radiation. $(\upsilon > \upsilon_0)$

(1) Only b and c (2) Only a, b & c

(3) a, b, c & d (4) Only a and c

19. An electron and a photon possess the same de-Broglie wavelength. If E_e and E_p respectively are the energies of electron and photon and v and c are their respective velocities, then $\frac{E_e}{E_p}$ is equal to

(1) $\frac{v}{c}$ (2) $\frac{v}{2c}$

(3) $\frac{v}{3c}$ (4) $\frac{v}{4c}$

20. The recoil momentum of H-atom due to the transition of an electron from $n = 4$ state to $n = 1$ state is

(1) $13.6\times10^{-19}\,kg\,ms^{-1}$

(2) $6.8\times10^{-27}\,kg\,ms^{-1}$

(3) $12.75\times10^{-247}\,kg\,ms^{-1}$

(4) None of these

21. Statement-1: Energies associated with nuclear processes are about a million times larger than energies in chemical process.

Statement-2: Energy released, when less tightly bound nuclei are transmuted into more tightly bound nuclei.

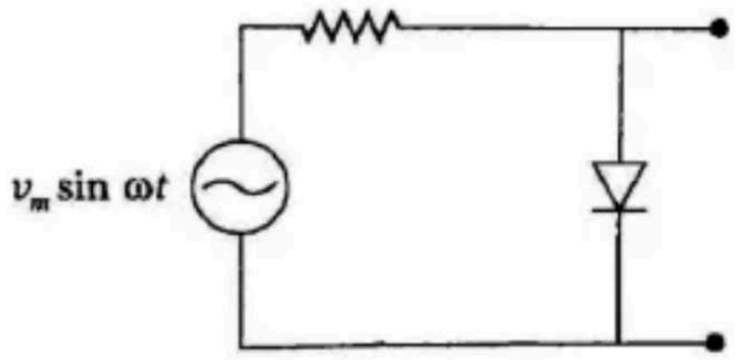
(1) Both statements are true.

(2) Statement-1 is true; statement-2 is false.

(3) Statement-1 is false; statement-2 is true.

(4) Both statements are false.

22. The output of the given circuit in figure given below,



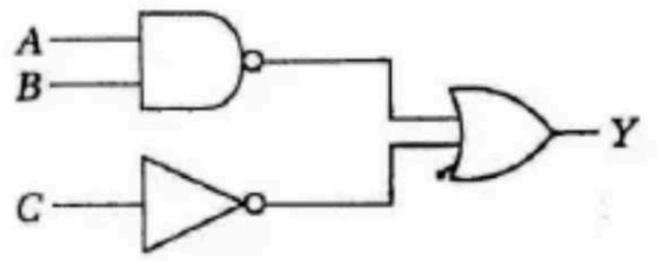
(1) would be zero at all times

(2) would be like a half wave rectifier with positive cycles in output

(3) would be like a half wave rectifier with negative cycles in output

(4) would be like that of a full wave rectifier

23. The inputs to the digital circuits are as shown below. The output Y is



(1) $A + B + \overline{C}$ (2) $(A + B)\overline{C}$

(3) $\overline{A} + \overline{B} + \overline{C}$ (4) $\overline{A} + \overline{B} + C$

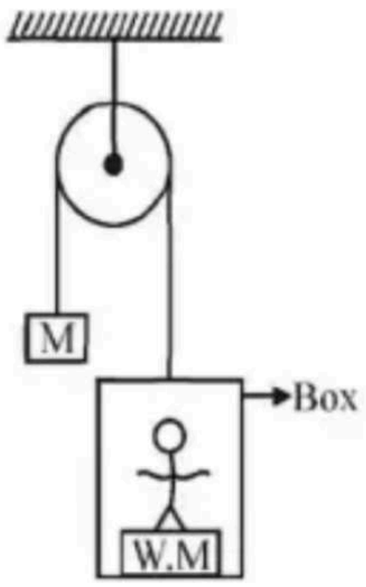
24. If no external voltage is applied across p-n junction, then there would be

(1) No electric field across the junction. (2) An electric field pointing from n-type to p-type side across the junction.

(3) An electric field pointing from p-type side across the junction.

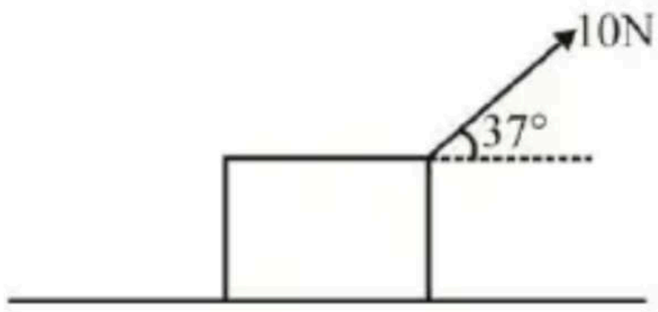
(4) A temporary electric field during formation of p-n junction that would subsequently disappear.

25. In the given diagram if mass of box is 20kg and mass of the man is 40kg. then the value of M for which reading of weighing machine is 480 N is (W.M=weighing machine)



(1) 60 kg (2) 90 kg
(3) 120 kg (4) 160 kg

26. A body of mass 5kg is at rest on horizontal table. The coefficient of friction between the body and the table is 0.6. A force of magnitude 10N is action on the block at an angle 37^0 from horizontal. The frictional force acting between the block and the table will be



(1) 30 N (2) 26.4 N
(3) 13.2 N (4) 8 N

27. Diameter of steel ball is measured using a Vernier calliper which has divisions of 0.1cm on it main scale (MS) and 10 divisions of its Vernier Scale (VS) match 9 divisions on main scale. Three such measurement for a ball are given as

S. No.	MS (cm)	VS Divisions
1	0.5	8
2	0.5	4
3	0.5	6

If the zero error is –0.03 cm, then the mean corrected diameter is

(1) 0.59 cm (2) 0.53 cm
(3) 0.58 cm (4) 0.56 cm

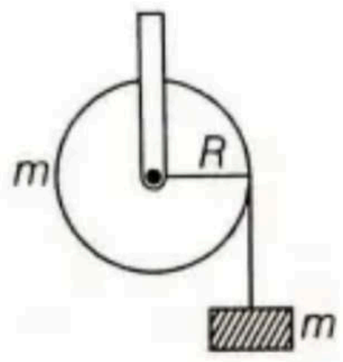
28. The equation of state of some gases can be expressed as $(P+\frac{a}{V^2})(V-b)=RT$, where P is the pressure, V is the Volume, T is the absolute temperature and a, b are constants the dimensions of 'a' is

(1) ML^5T^{-2} (2) $ML^{-1}T^{-2}$
(3) L^3 (4) L^6

29. A particle moves on a straight line with a uniform velocity. The angular momentum of the particle is
(i) Always zero
(ii) Zero about a point on the straight line
(iii) Zero about a point not lies on the straight line
(iv) Constant always about a given point, not on the line

(1) (i, ii) (2) (ii, iii)
(3) (i, iv) (4) (ii, iv)

30. A mass m supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release?



(1) $\frac{2g}{3}$ (2) $\frac{g}{2}$
(3) $\frac{5g}{6}$ (4) g

31. A Scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and comes to rest. The correct value of t_1/t_2 will be

(1) $\frac{a_1+a_2}{a_2}$ (2) $\frac{a_2}{a_1}$
(3) $\frac{a_1}{a_2}$ (4) $\frac{a_1+a_2}{a_1}$

32. A particle is moving eastwards with a velocity $5ms^{-1}$. In 10 second, the velocity changes to $5ms^{-1}$ northwards. The average acceleration in this time is

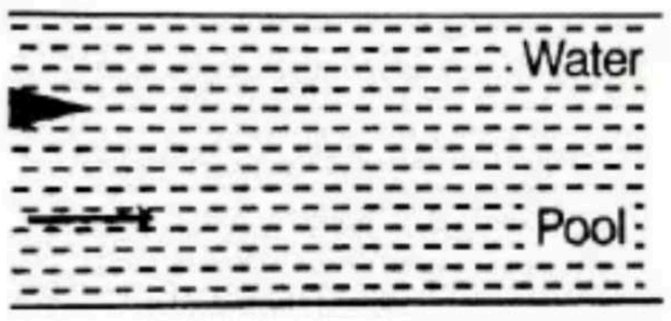
(1) $\frac{1}{\sqrt{2}} ms^{-2}$ towards North-East

(2) $\frac{1}{2} ms^{-2}$ towards North

(3) Zero

(4) $\frac{1}{\sqrt{2}} ms^{-2}$ towards North-West

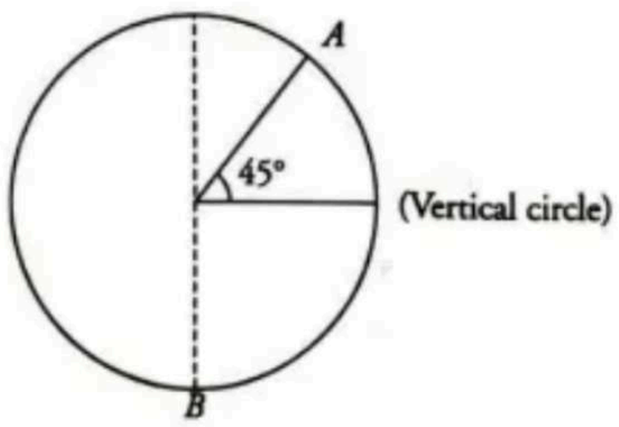
33. A bullet of mass 200 g having initial kinetic energy 90 J is shot inside a long swimming pool as shown in the figure. If its kinetic energy reduces to 40 J within 1 second, the minimum length of the pool, the bullet has to travel, so that it completely comes to rest is



(1) 45 m (2) 90 m

(3) 125 m (4) 25 m

34. A body of m kg slides from rest along the curve of vertical circle from point A to B in frictionless path. The velocity of the body at B is



(given, R=14m, $g = 10m/s^2$ and $\sqrt{2} = 1.4$)

(1) 19.8 m/s (2) 16.7 m/s

(3) 21.9 m/s (4) 10.6 m/s

35. Assertion: Earth has atmosphere whereas moon doesn't have any atmosphere.

Reason: The escape velocity on moon is very small as compared to that on earth.

(1) Both Assertion and Reason are True, and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True, but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False, but Reason is True.

36. A satellite is moving with a constant speed 'v' in circular orbit around the earth. An object of mass *'m'* is ejected from the satellite such that it just escapes from the gravitation pull of the earth. At the time of ejection, the kinetic energy of the object is

(1) $\frac{3}{2}mv^2$ (2) $2mv^2$

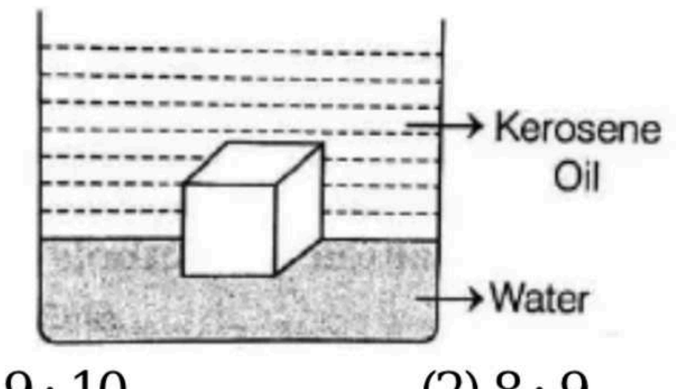
(3) mv^2 (4) $\frac{1}{2}mv^2$

37. The elastic limit of brass is 379 MPa. What should be the minimum diameter of a brass wire, if it is to supports a 400 N load without exceeding its elastic limit?

(1) 0.90 mm (2) 1.00 mm

(3) 1.15 mm (4) 1.36 mm

38. A cube of ice floats partly in water and partly in kerosene oil. The ratio of volume of ice immersed in water to that in kerosene oil (specific gravity of kerosene oil = 0.8, specific gravity of ice =0.9)



(1) 9 : 10 (2) 8 : 9

(3) 5 : 4 (4) 1 : 1

39. The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is

(1) 1 : 9 (2) 1 : 81

(3) 1 : 3 (4) 1 : 27

40. Two different wires having lengths L_1 and L_2 and respective temperature coefficients of linear expansion α_1 and α_2, are joined end-to-end. Then the effective temperature coefficient of linear expansion is

(1) $\frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$

(2) $2\sqrt{\alpha_1 \alpha_2}$

(3) $\frac{\alpha_1 + \alpha_2}{2}$

(4) $4\frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}\frac{L_2 L_1}{(L_2 + L_1)^2}$

41. Two materials having coefficients of thermal conductivity 3K and K and thickness d and 3d respectively, are joined to form a slab. The temperature of the outer surfaces are θ_2 and θ_1 respectively, ($\theta_2 > \theta_1$). The temperature at the interface is

(1) $\frac{\theta_2 + \theta_1}{2}$ (2) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

(3) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$ (4) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$

42. A litre of dry air at STP expands adiabatically to a volume of 3L. If $\gamma = 1.40$, then work done air is ($3^{1.4} = 4.6555$) [Take, air to be an ideal gas]

(1) 100.8 J (2) 90.5 J

(3) 48 J (4) 60.7 J

43. A particle executes SHM of amplitude A. The distance from the mean position when it's kinetic energy becomes equal to its potential energy is

(1) $\frac{1}{\sqrt{2}}$ A (2) $\sqrt{2A}$

(3) $\frac{1}{2}$A (4) 2A

44. A pipe open at both ends has a fundamental frequency f in air. The pipe is dipped vertically in water, so that half of it is in water. The fundamental frequency of the air column is now

(1) $\frac{f}{2}$ (2) $\frac{3f}{4}$

(3) 2f (4) f

45. The relation between root mean square speed v_{rms} and most probable speed v_p for the molar mass M of oxygen gas molecule at the temperature of 300 K will be

(1) $v_{rms} = \sqrt{\frac{2}{3}}v_p$ (2) $v_{rms} = \sqrt{\frac{3}{2}}v_p$

(3) $v_{rms} = v_p$ (4) $v_{rms} = \sqrt{\frac{1}{3}}v_p$

CHEMISTRY

46. The ratio of wavelengths of electron wave in the first four orbits of hydrogen is

(1) 1 : 2 : 3 : 4 (2) $1 : \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$

(3) 1 : 4 : 9 : 16 (4) $1 : \frac{1}{4} : \frac{1}{9} : \frac{1}{16}$

47. Which one of the following statement is incorrect ?

(1) Atomic mass of Calcium is 40u

(2) Molar mass of Oxygen is 32u

(3) Atomic weight of Magnesium is 24u

(4) Gram molecular weight of Hydrogen is 2g

48. Match the following

Column - I (Element)	Column – II (Oxidation number in its compounds)
a) Potassium	i) – 1
b) Barium	ii) +3
c) Fluorine	iii) +1
d) Aluminium	iv) +2

Correct match is

	a	b	c	d
(1)	iii	ii	iv	i
(2)	iii	i	ii	iv
(3)	iii	ii	i	iv
(4)	iii	iv	i	ii

49. Which one of the following pair is isoelectronic and isostructural ?

(1) NH_3, H_2O (2) SiH_4, NH_4^+

(3) NH_3, CH_3^- (4) H_2S, NH_2^-

50. Statement – I : N – F bond has less electronegativity difference than N – H bond.
Statement- II: Dipole moment of $NF_3 < NH_3$

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

51. Correct statements regarding hybridization are

A) Orbitals of different atoms mix with each other to form hybrid orbitals.

B) Vacant orbitals cannot involve in hybridization.

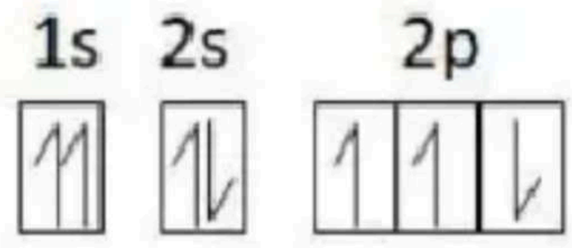
C) Each hybrid orbital can accommodate a maximum of two electrons with opposite spin.

D) Hybrid orbitals cannot form sigma bonds.

(1) A, B, C and D (2) C only

(3) C and D only (4) A and C only

52. Consider the following electronic configuration



This configuration violates

(1) Hund's rule and Pauli's principle only

(2) Pauli's principle and Aufbau's principle only

(3) Hund's rule only

(4) Aufbau's principle, Hund's rule and Pauli's principle

53. During the qualitative analysis of salt with cation y^{2+}, addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation $\left(y^{2+}\right)$ present respectively are

(1) Dimethylglyoxime and Ni^{2+}

(2) Dimethylglyoxime and Co^{2+}

(3) Nessler's reagent and Hg^{2+}

(4) Nessler's reagent and Ni^{2+}

54. In the flame test of a mixture of salts, a green flame with blue centre was observed. Which one of following cations may be present ?

(1) Cu^{2+} (2) Sr^{2+}

(3) Ba^{2+} (4) Ca^{2+}

55. Solute A dimerises in water. When 0.7g of solute A is dissolved in 42.0g water, it depresses the freezing point by $0.2^{\circ}C$. The percentage association of solute A in water is
(Given : Molar mass of $A = 93 g\,mol^{-1}$. Molal depression constant of water is $1.86 K\,kg\,mol^{-1}$)

(1) 50 % (2) 60 %

(3) 70 % (4) 80 %

56. At 345K, the half life for the decomposition of a sample of a gaseous compound initially at 55.5kPa was 340s. When the pressure was 27.75kPa, the half life was found to be 170s. The order of reaction is

(1) 0 (2) 1

(3) 2 (4) 3

57. Which one of the following is a water soluble vitamin, that is not excreted easily?

(1) Vitamin B_2 (2) Vitamin B_1

(3) Vitamin B_6 (4) Vitamin B_{12}

58. At 298 K, the equilibrium constant (K) for a reaction is 10^5. The value of standard Gibbs free energy change (ΔG°) is (R = 8.314 J $mol^{-1} K^{-1}$):

(1) $-28.5 kJmol^{-1}$ (2) $-11.4 kJmol^{-1}$

(3) $+28.5 kJmol^{-1}$ (4) $+11.4\ kJ\ mol^{-1}$

59. Which among the following pair are globular proteins ?

(1) Keratin and insulin

(2) Keratin and albumins

(3) Keratin and myosin

(4) Insulin and albumins

60. P^H of which of the following aqueous solution will have highest value ?

(1) NaCl (2) CH_3COONH_4

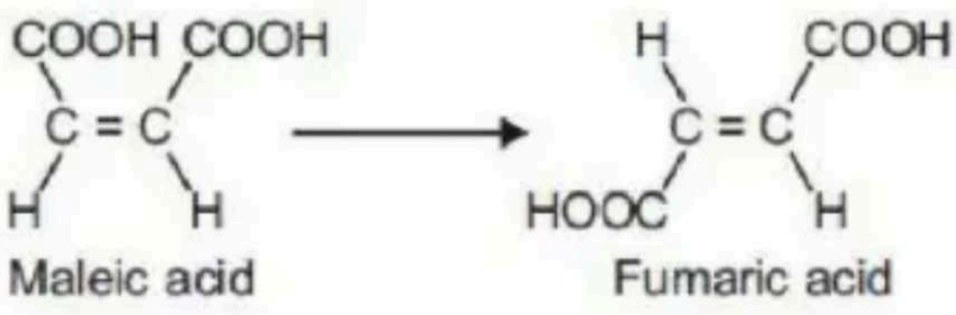
(3) NH_4CN (4) CH_3COONa

61. If one mole of N_2O_4 decomposes to 60% into NO_2 , then find its Kc value in a one litre vessel at 298K

(1) 7.5 (2) 3.6

(3) 1.5 (4) 4.9

62. The standard enthalpies of combustion of fumaric acid and maleic acid are $-1336.0 kJ\ mol^{-1}$ and $-1359.2 kJ/mol$ respectively. What will be the enthalpy of the given isomerisation process ?



(1) Zero (2) $+23.2 kJ/mol$

(3) $-23.2 kJ/mol$ (4) $+46.4 kJ/mol$

63. A copper electrode is dipped in a solution of its salt at 298 K. The effective concentration of Cu^{2+} ions in the solution is 0.1 M. Calculate the electrode potential. $\left[E^o_{Cu^{+2}/Cu} = +0.34V\right]$

(1) +0.301 V (2) +0.34 V

(3) +0.31 V (4) +0.37 V

64. At 298K, if a strong acid solution of $P^H = 6$ is diluted by 100 times then the P^H of resulting solution is

(1) 8 (2) 6

(3) 6.96 (4) 6.3

65. The following results have been obtained during the kinetic studies of the reaction

$2P + Q \longrightarrow R + S$

Experiment	[P] mol/lit	[Q] mol/lit	Rate of reaction (mol/lit/sec)
I	0.1	0.1	6×10^{-3}
II	0.4	0.1	2.4×10^{-2}
III	0.4	0.2	9.6×10^{-2}
IV	0.8	0.4	76.8×10^{-2}

The correct option for the rate equation for above reaction is

(1) $Rate = K[P]^2[Q]$

(2) $Rate = K[P]^2[Q]^o$

(3) $Rate = K[P][Q]$

(4) $Rate = K[P]^1[Q]^2$

66. During the discharge of lead storage battery, which of the following does not occur

(1) $PbSO_4$ is formed

(2) H_2SO_4 is consumed

(3) Voltage of the cell increases

(4) PbO_2 gets reduced

67. 31 grams of white phosphorous(P_4) when burnt in excess of oxygen at 1 atmosphere liberated 'Q' kJ of heat. Enthalpy of combustion of white phosphorous is ___ (kJ/mole)

(1) $-4Q$ (2) $-Q$

(3) $-2Q$ (4) $-Q/4$

68. 54ml of liquid 'A' is mixed with 54ml of liquid 'B'. The volume of the mixture formed is 108ml. The two liquids are most likely to be

(1) n – hexane +n – heptane

(2) Acetone + Carbon tetrachloride

(3) Ethanol + Cyclo hexane

(4) Hydrochloric acid + water

69. 0.12 g of an organic compound gave 0.222g $Mg_2P_2O_7$ by the usual analysis. The weight percentage of phosphorus in the compound is nearly

(1) 15.23 (2) 38.75

(3) 51.66 (4) 60.92

70.

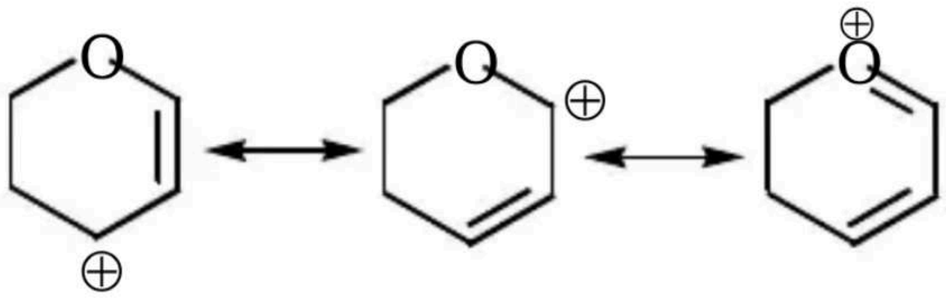
$$CO_2 + C_6H_5MgBr \xrightarrow[H_3O^+]{\text{Dry ether}} A \xrightarrow[\Delta]{NH_3} B \xrightarrow{Br_2+KOH} C \xrightarrow[II)\,H_2O,283K]{I)\,NaNO_2+HCl/278K} D \xrightarrow{Zn} E$$

Here 'E' is

(1) Benzene (2) Phenol

(3) Aniline (4) Benzamide

71. The most stable canonical structure among the following is



I II III

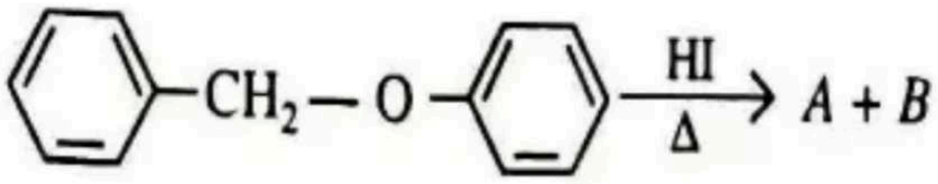
(1) I

(2) II

(3) III

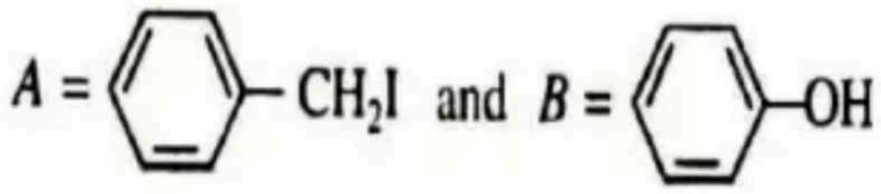
(4) All have equal stability.

72. Consider the following reaction:

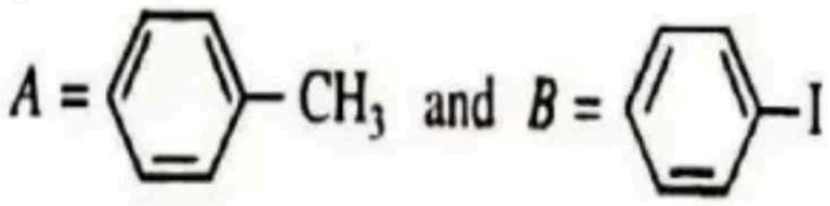


Identify products A and B.

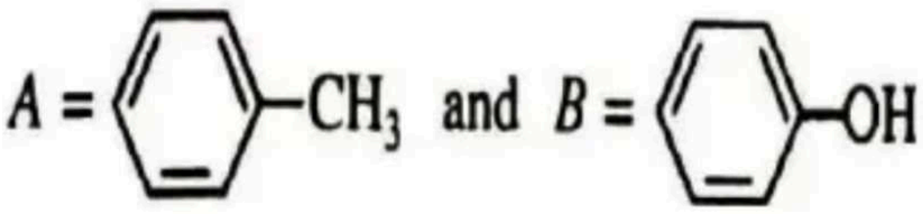
(1)



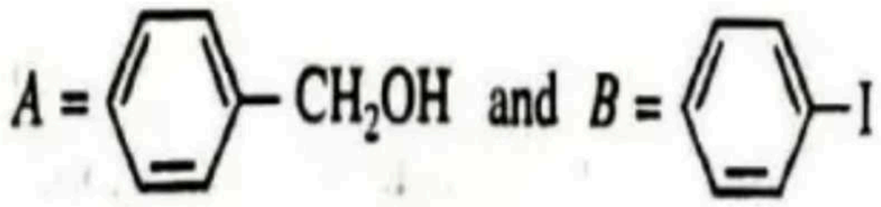
(2)



(3)



(4)



73. Match the Column I with Column II.

Column – I (Reactions)	Column–II (products)
A) Benzene + Cl_2 $\xrightarrow{AlCl_3}$	I) Benzoic acid
B) Benzene + $CH_3Cl \xrightarrow{AlCl_3}$	II) Methyl phenyl ketone
C) Benzene + $CH_3COCl \xrightarrow{AlCl_3}$	III) Toluene
D) Toluene $\xrightarrow[H^+]{KMnO_4/KOH}$	IV) Chloro benzene
	V) Benzene hexachloride

(1) A – II, B – I, C – III, D – IV

(2) A – II, B – I, C – III, D – IV

(3) A – III, B – IV, C – I, D – II

(4) A – IV, B – III, C – II, D – I

74. In which of the following pair of oxides both are amphoteric

(1) CrO_3, Cr_2O_3 (2) CrO_3, Mn_2O_7

(3) Cr_2O_3, V_2O_5 (4) V_2O_5, Mn_2O_7

75. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : $C_6H_5N_2^+Cl^-$ is more stable than $C_2H_5N_2^+Cl^-$

Reason (R) : $C_6H_5N_2^+$ is resonance stabilized, whereas $C_2H_5N_2^+$ is not resonance stabilized.

Choose the correct answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

76. Correct statements among the following

A) XeO_3 has a pyramidal molecular structure

B) $XeOF_4$ and has a square pyramidal molecular structure

C) XeO_4 has tetrahedral structure

D) Ne is the least abundant gas in air among the group-18 elements.

(1) A, B and C only

(2) B and C only

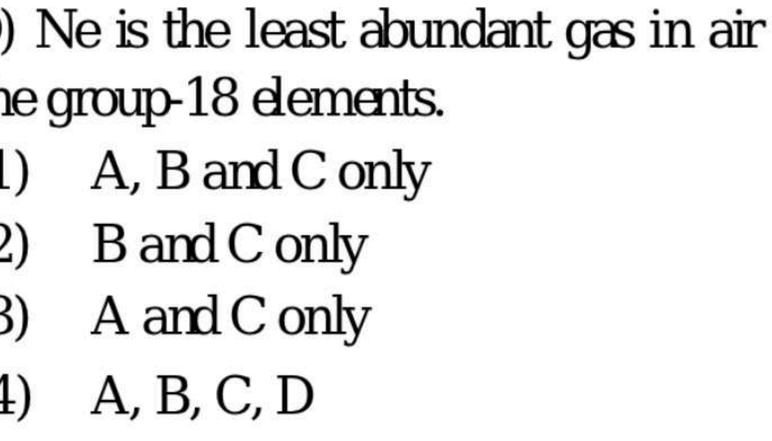
(3) A and C only

(4) A, B, C, D

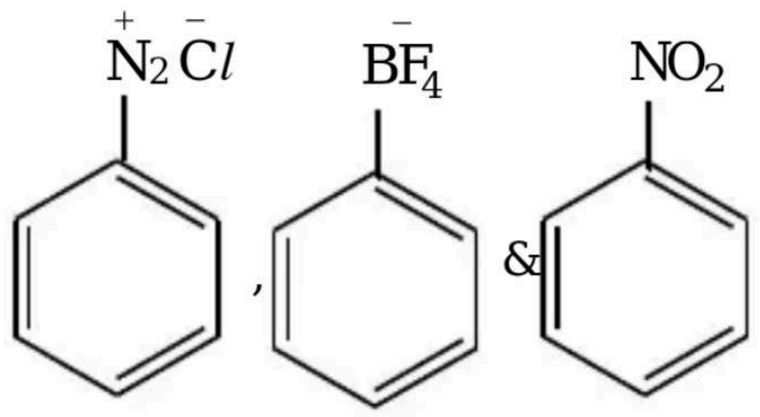
77. $C_6H_5NH_2 \xrightarrow[0-5^0C]{HNO_2+HCl} X \xrightarrow{HBF_4} Y \xrightarrow[Cu/\Delta]{NaNO_2} Z$

In this sequence X, Y and Z are respectively

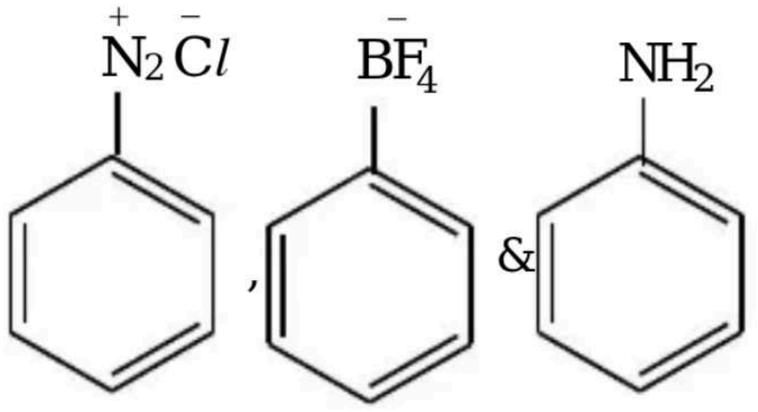
(1)



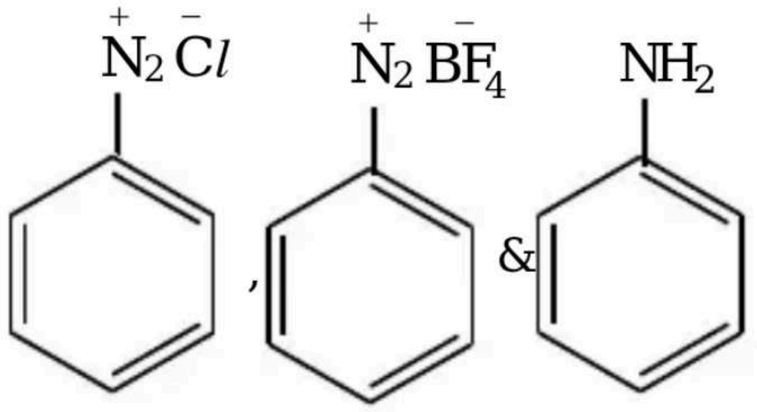
(2)



(3)



(4)



78. Optically active complexes among the following are

(A) $cis-\left[CoCl_2(en)_2\right]Cl$

(B) $trans-\left[CoCl_2(en)_2\right]Cl$

(C) $\left[Co(NH_3)_3Cl_3\right]$

(D) $\left[Cr(en)_3\right]Cl_3$

(1) A, B and D only (2) B, C and D only

(3) C & D only (4) A & D only

79. $CaC_2 \xrightarrow{H_2O} A \xrightarrow{Red\ hot\ Cu\ tube} B \xrightarrow[CH_3Cl]{AlCl_3} C$

In the above sequence, the reaction $A \rightarrow B$ is _____ and reaction $B \rightarrow C$ is _____

(1) Polymerisation, free radical substitution

(2) Polymerisation, nucleophilic substitution

(3)Polymerisation, electrophilic substitution

(4) Isomerisation, electrophilic addition

80. The incorrect statement among the following is

A) The first ionization enthalpies of Lanthanoids is nearly 600 kJ/mole

B) Highest oxidation state exhibited by Neptunium in its compounds is '+7'

C) The transition metal present in Ziegler – Natta catalyst is Titanium

D) Europium has smallest atomic radius among lanthanoids

(1) Both A and B (2) Both C and D

(3) Only C (4) Only D

81. Match the following

Column I (Example)	Column II (Reaction)
A) $CH_3COCl \xrightarrow{H_2/Pd-C/BaSO_4} CH_3CHO$	i) HVZ reaction
B) $CH_3-CHO \xrightarrow{dil.\ NaOH} CH_3-CH=CH-CHO$	ii) Rosenmund reaction
C) $CH_3CH_2COOH \xrightarrow[RedP]{Br_2} CH_3-CHBr-COOH$	iii) Clemenson reduction
D) $CH_3CHO \xrightarrow[HCl]{Zn-Hg} CH_3-CH_3$	iv) Aldol condensation

The correct combination is

(1) A-i, B-ii, C-iii, D-iv

(2) A-iii, B-iv, C-i, D-ii

(3) A-ii, B-iv, C-i, D-iii

(4) A-iv, B-ii, C-i, D-iii

82. With respect to the conformers of ethane, which of the following statements is true?

(1) Bond angle changes but bond length remains same

(2) Both bond angle and bond length change

(3) Both bond angle and bond length remain same

(4) Bond angle remains same but bond length changes

83. The element with atomic number 115 belongs to which family and what is its electronic configuration.

(1) Halogen family, $[Rn]\,5f^{14}\,6d^{10}\,7s^2\,7p^5$

(2) Carbon family, $[Rn]\,5f^{14}\,6d^{10}\,7s^2\,7p^2$

(3) Oxygen family, $[Rn]\,5f^{14}\,6d^{10}\,7s^2\,p^4$

(4) Nitrogen family, $[Rn]5f^{14}\,6d^{10}7s^2 7p^3$

84. Match Column-I with Column -II :

	Column-I (Reactions)		Column-II (Reagents)
A.	Ethanenitrile to ethanal	p.	PCC
B.	Allyl alcohol to propenal	q.	DIBAL-H
C.	But-2-ene to ethanal	r.	O_3Zn/H_2O

Choose the correct answer from the options given below :

(1) A-q, B-p, C-r (2) A-r, B-q, C- p

(3) A-r, B- p, C-q (4) A- q, B-r, C- p

85. Which of the following arrangements represents correct order of least negative to most negative electron gain enthalpy for C, Ca, Al, F and O ?

(1) Ca <Al <C <O <F

(2) Al <Ca <O <C <F

(3) Al <O <C <Ca <F

(4) C <F <O <Al <Ca

86. $CH_3-CH_2-CH=CH_2(X) \xrightarrow[(C_6H_5CO)_2O_2]{HBr} A \xrightarrow[b)HBr]{a)alc.KOH/\Delta} B \xrightarrow[c)NaNH_2/\Delta]{a)alc.KOH/\Delta,\ b)Br_2/CCl_4} C.$

(I) A and B are same compounds

(II) 'C' contains acidic hydrogen atoms

(III) 'C' on hydrogenation in presence of Lindlar's catalyst gives positional isomer of X

(IV) Both 'A' and 'B' give same product when reduced with Zn/HCl

Identify whether the above statements are true (T) or false (F)

(1) TTFF (2) FFTF

(3) TFTT (4) FFTT

87. Chlorobenzene $\xrightarrow{X}$ Phenol $\xrightarrow{Y}$ salicylaldehyde

The reactions 'X' and 'Y' are respectively

(1) Fittig and Kolbes

(2) Swart's and Reimer-Tiemann

(3) Aromatic nucleophilic substitution and Reimer - Tiemann

(4) Fittig and Friedel craft

88. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :

Assertion (A) : $\left[Co(C_2O_4)_3\right]^{-3}$ complex is an inner orbital complex and is diamagnetic.

Reason (R) : $C_2O_4^{-2}$ is strong field ligand in Co^{+3} octahedral complexes.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is incorrect but R is correct

89. In 13th group, the element with least melting point is

(1) B (2) Al

(3) Ga (4) In

90. $\underset{(I)}{(C_6H_5)_2C(Br)(CH_3)}$,

$\underset{(II)}{(C_6H_5)CH(Br)(CH_3)}$,

$\underset{(III)}{,(C_6H_5)C(Br)(CH_3)_2}$

The order of decreasing S_N1 reactivity is

(1) I >II >III (2) I >III >II

(3) II >III >I (4) III >I >II

BIOLOGY

91. Select the *incorrect* option.

(1) Mammary alveoli secrete milk.

(2) The mammary ampulla stores milk temporarily.

(3) Each breast contains 15 to 20 mammary lobes.

(4) Lactiferous ducts connect mammary tubules with mammary ducts

92. The taxonomic categories having the group of related genera are

(1) Sapindales; Polymoniales

(2) Anacardiaceae; Primata

(3) Diptera; Primata

(4) Canidae; Solanaceae

93. A woman is infertile due to thick and tough zona pellucida developed around the secondary oocytes she produces. Suggest a suitable ART procedure to solve the problem.

(1) Artificial Insemination

(2) GIFT

(3) ICSI

(4) IUI

94. Read the following statements and select the correct option.

Statement-I: Few plants are partially heterotrophic such as insectivorous plants or parasites.

Statement-II: In Whittaker's classification (five kingdom classification) there is no mention of Lichens, Viruses, Viroids and Prions.

(1) Statement-I is correct, but Statement-II is incorrect.

(2) Statement-I is incorrect, but Statement-II is correct.

(3) Both Statement-I and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

95. The chances of *Toxoplasma* infection usually increase in patients suffering from
(1) AIDS (2) amoebiasis
(3) dengue (4) hepatitis

96. The main contraceptive actions of hormone releasing IUDs do not include
(1) phagocytosis of sperm.
(2) making the uterus unsuitable for implantation.
(3) making the cervix hostile to sperm.
(4) prevention of ovulation

97. Based on the differentiation of external genital organs, what is the earliest stage of gestation at which the sex of the developing foetus can be reliably identified using ultrasound?
(1) End of 12th week
(2) End of 20th week
(3) Beginning of 24th week
(4) Just after the implantation

98. Which of the following is incorrectly matched?
(1) *Ulothrix* – Mannitol
(2) *Volvox* - Starch
(3) *Ectocarpus* – Fucoxanthin
(4) *Porphyra* – Floridean starch

99. Assertion(A): Micro and mega sporophylls are born on different plants in *Cycas*.
Reason(R): *Cycas* is a dioecious gymnosperm.
(1) Both (A) and (R) are correct, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) Both (A) and (R) are false statements.

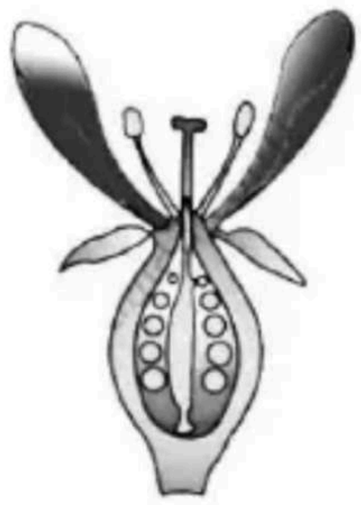
100. In which of the following families, the members are mostly with monoadelphous stamens and anthers are monothecous.
(1) Fabaceae (2) Malvaceae
(3) Brassicaceae (4) Liliaceae

101. Read the following.

A. Poikilotherm	B. Shelled eggs
C. Homeotherm	D. Tetrapod
E. Ureotelic	F. Viviparity

Which of the above features are applicable to frogs?
(1) A, B, and C (2) A, B and F
(3) A, D, and E (4) B, C and E

102. Which of the following statements is incorrect with respect to the following figure.



(1) The margin of thalamus grows upwards enclosing the ovary completely and getting fused with it.
(2) Ovary is said to be inferior.
(3) The flower is said to be perigynous.
(4) It is seen in the flowers of guava and cucumber.

103. Match the following.

	List I (Scientific name)		List II (Common name)
A	*Spongilla*	I.	Sea walnut
B	*Meandrina*	II.	Sea anemone
C	*Adamsia*	III.	Brain coral
D	*Ctenoplana*	IV.	Fresh water sponge
		V.	Sea cucumber

(1) A-IV, B-V, C-II, D-I
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-IV, B-III, C-II, D-V

104. Choose the correct option of the following

Statement-I: Potato; ginger; turmeric; Zaminkand and *Colocasia* are modified to store food in their underground stems.

Statement-II: They are the organs of perennation to tide over unfavourable conditions for growth.

(1) Statement I is correct, but statement II is incorrect.

(2) Statement I is incorrect, but statement II is correct.

(3) Both statement I and statement II are correct.

(4) Both statement I and statement II are incorrect.

105. Which of the following features is shared by grasshopper, *Saccoglossus* and *Chaetopleura*?

(1) Dorsal nerve cord

(2) Segmented body

(3) Open type of circulation

(4) Direct development

106. The pericycle of one of the following initiates not only lateral roots but also vascular cambium during secondary growth.

(1) Dicot stem (2) Dicot root

(3) Monocot stem (4) Monocot root

107. Identify the animals A, B and C based on the description below.

A. Electric organ

B. Pneumatic bones

C. Oviparous mammal

Select the correct option.

	A	B	C
(1)	*Torpedo*	*Neophron*	*Ornithorhynchus*
(2)	*Trygon*	*Apteynodytes*	*Balaenoptera*
(3)	*Vipera*	*Corvus*	*Pteropus*
(4)	*Pristis*	*Struthio*	*Equus*

108. Pollen grains can be stored for several years in liquid nitrogen having temperature of

(1) -196°C (2) -80°C

(3) -120°C (4) -160°C

109. Given below are the characteristics of flowers, how many of them are associated with flowers pollinated by insects ?

Brightly coloured, Presence of nectaries, Sticky pollens, Feathery stigmas, Foul odour, Winged pollen, Mucilaginous covering around pollen grain.

(1) Four (2) Three

(3) Five (4) Six

110. Given below are two statements.

Statement I: Cell junctions provide both structural and functional links between individual cells of a tissue.

Statement II: Cells in epithelium are held together with little intercellular material.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are incorrect.

(2) Statement I is incorrect, but statement II is correct.

(3) Statement I is correct, but statement II is incorrect.

(4) Both statement I and statement II are correct

111. Any plane passing through the central axis of the body divides the organism into two identical halves, it is called

(1) bilateral symmetry

(2) asymmetry

(3) radial symmetry

(4) biradial symmetry

112. Match the following and select the correct option.

Column-I		Column-II	
A	Uridine	I	Nitrogen base
B	Adenylic acid	II	Nucleoside
C	Pyrimidine	III	Nucleic acid
D	Ribonucleic acid	IV	Nucleotide

(1) A-IV, B-II, C-I, D-III

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-I, B-III, C-IV, D-II

113. The patella is composed of which tissue?
(1) Cartilage (2) Muscle
(3) Ligament (4) Bone

114. Who first identified DNA as an acidic substance in the nucleus and named it 'Nuclein'?
(1) Friedrich Miescher.
(2) Rosalind Franklin.
(3) Erwin Chargaff
(4) James Watson

115. Arrange the following formed elements in ascending order of their abundance in blood and choose the correct option.
(A) Eosinophils (B) Monocytes
(C) Basophils (D) Neutrophils
(E) Lymphocytes
(1) C-B-A-E-D (2) C-B-E-A-D
(3) C-A-B-E-D (4) D-E-B-A-C

116. Assertion(A): Amino acids are called α-amino acids.
Reason(R): In the structure of an amino acid, an amino group and a carboxyl group are substituents on the same carbon.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false, but (R) is true.

117. Read the following statements and choose the correct option.
Statement-I: The sequence of amino acids i.e., positional information in a protein is shown by its primary structure.
Statement-II: Only Quaternary structures are responsible for many biological activities of proteins.
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct, but statement II is incorrect.
(4) Statement I is incorrect, but statement II is correct.

118. Identify vector-borne diseases from the following.
A. Chikungunya B. Dengue
C. Malaria D. Amoebiasis
E. Filariasis F. Ascariasis
Choose the correct option.
(1) A, B, and F only
(2) A, B, C and E
(3) B, D, and F only
(4) A, C and D only

119. Cellular oncogenes (proto-oncogenes) are
(1) found only in cancer cells
(2) absent in normal cells
(3) found only in viruses
(4) present in normal cells

120. Antibodies that cross the placenta provide
(1) artificial active immunity
(2) natural passive immunity
(3) artificial passive immunity
(4) innate immunity

121. Match Column-I with Column-II

Column-I		Column-II	
A	Cristate	I	Primary constriction of a chromosome
B	Thylakoids	II	Disc shaped sacs in Golgi complex
C	Centromere	III	Infoldings of inner mitochondrial membrane
D	Cisternae	IV	Flattened sacs in stroma of plastid

A B C D
(1) III IV I II
(2) III I II IV
(3) II III I IV
(4) II IV I III

122. Match the following.

	Disease		Symptom
A.	Ascariasis	I.	Inflammation of lymphatic vessels
B.	Filariasis	II.	Internal bleeding, fever, blockage of intestinal passage
C.	Dengue	III.	Dry, scaly lesions on skin, nail, scalp etc.
D.	Ringworm	IV.	Joint pains, retro-orbital pain (pain behind eyes)

(1) A-II, B-IV, C-III, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-II, C-I, D-III

123. Choose the *incorrect* statement from the following

(1) As per fluid mosaic model of plasma membrane, the non polar tails of saturated hydrocarbons is protected from the aqueous environment.

(2) Only '80S' ribosomes are composed of two sub units.

(3) Both centrioles in a centrosome lie perpendicular to each other.

(4) Nucleolus is a site of active ribosomal RNA synthesis.

124. Read the following statements regarding predation and select the *incorrect* statements.

A. Predators act as conduits for energy transfer across trophic levels.

B. Predators always increase the intensity of competition among competing prey species.

C. Predators are not prudent.

D. Predators help in maintaining species diversity in a community.

(1) B and D (2) B and C

(3) A and C (4) A and D

125. Assertion(A): In case of acrocentric chromosome the centromere is situated slightly away from the middle of the chromosome.

Reason(R): Kinetochores are the structures on the sides of centromere where spindle fibres are attached during metaphase.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

126. Which of the following statements about a pond ecosystem is *incorrect*?

(1) There is bidirectional movement of energy towards the higher trophic levels and its return to the producers.

(2) The solar input, the cycle of temperature, day-length and other climatic conditions regulate the rate of function of the entire pond.

(3) The pond ecosystem performs conversion of inorganic into organic material with the help of the radiant energy of the sun by the autotrophs.

(4) The decomposers of the pond ecosystem are the fungi, bacteria and flagellates especially abundant in the bottom of the pond

127. Arrange the following events of meiosis in a correct sequence

A) Terminalization

B) Crossing over

C) Synapsis

D) Homologous chromosomes separation (Disjunction)

(1) C, B, D, A (2) C, B, A, D

(3) B, A, C, D (4) C, D, B, A

128. Read the following statements.

Statement I: All natural ecosystems are self-sustainable entities.

Statement II: In any natural ecosystem, complex interactions exist.

(1) Both statement I and statement II are incorrect.

(2) Statement I is incorrect, but statement II is correct.

(3) Statement I is correct, but statement II is incorrect.

(4) Both statement I and statement II are correct

129. In meiosis I, a bivalent is an association of

(1) Four chromatids and four centromeres.

(2) Two chromatids and two centromeres.

(3) Two chromatids and one centromere.

(4) Four chromatids and two centromeres.

130. Which of the following is an example of man-made terrestrial ecosystem?

(1) Forest (2) Grassland

(3) Desert (4) Crop field

131. Assertion (A): Alexander von Humboldt found that within a region, species richness increased with increasing explored area, but only up to a limit.

Reason (R): Relationship between species richness (S) and area (A) for a wide variety of taxa turns out to be a rectangular hyperbola on a logarithmic scale.

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is NOT the correct explanation of A.

(3) A is not correct but R is correct.

(4) A is correct but R is not correct

132. A plant cell has 16 bivalents in Prophase-I, of meiosis. How many chromosomes and chromatids will be present in each gamete of this plant respectively?

(1) 8 chromosomes and 8 chromatids

(2) 16 chromosomes and 12 chromatids

(3) 16 chromosomes and 16 chromatids

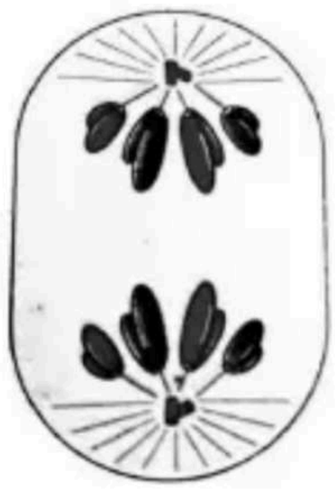
(4) 8 chromosomes and 16 chromatids

133. Which of the following is *not* an example of a narrowly utilitarian argument for the conservation of biodiversity?

(1) Medicines (2) Resins

(3) Pollination (4) Firewood

134. Identify the stage of the cell division and select the correct option.



(1) It represents the beginning of nuclear division.

(2) Centromeres split and chromatids separate.

(3) Micro tubules of spindle fibres get attached to the kinetochore.

(4) Nuclear membrane and nucleolus reappear in the cell.

135. Which of the following statements about biodiversity hotspots is *incorrect*?

(1) They are regions with very high levels of species richness.

(2) They are regions of accelerated habitat loss.

(3) They are regions with high degree of endemism.

(4) All the biodiversity hotspots put together cover more than 2 per cent of the Earth's area

136. Match the following

Column-I		Column-II	
A	Citric acid	1	Act as competitive inhibitor
B	Statins	2	Used in organ-transplant patients
C	Streptokinase	3	Produced by *Aspergillus*
D	Cyclosporin A	4	Used in patients, undergone myocardial infarction

(1) A – 1, B – 2, C – 3, D – 4

(2) A – 3, B – 4, C – 2, D – 1

(3) A – 3, B – 1, C – 4, D – 2

(4) A – 3, B – 4, C – 1, D – 2

137. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Drones of honeybees do not have a father but have a mother.

Reason (R) : Drones of honeybees develop from unfertilised eggs through a process called parthenogenesis.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false

138. Statement-I: PSI and PSII are named in sequence of their function during the light reactions (or) photo chemical phase.

Statement-II: Each photosystem has only one molecule of chlorophyll-a, forming a light harvesting complex called antenna.

(1) Both statement I and statement II are true.

(2) Both statement I and statement II are false.

(3) Statement I is true, but statement II is false.

(4) Statement I is false, but statement II is true.

139. The levels of which of the following hormones increase during follicular phase of menstrual cycle?

(1) GnRH and oestrogen

(2) FSH and progesterone

(3) LH and progesterone

(4) Oestrogen and progesterone

140. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : If both parents are heterozygous carriers of sickle-cell anaemia, their offspring may express the disorder.

Reason (R) : Sickel-cell anaemia is an autosomal dominant disorder.

In light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is false, but (R) is true.

(4) (A) is true, but (R) is false

141. Which of the following is not true regarding ATP synthesis through chemiosmosis in chloroplast?

(1) Movement of protons across the thylakoid membrane to stroma.

(2) Reduction of NADP to $NADPH_2$ on the stroma side of the membrane.

(3) Break down of proton gradient.

(4) Break down of electron gradient.

142. Statement I: Polymorphisms in DNA are inheritable from parents to children.
Statement II: If a mutation in a germ cell does not affect reproduction, it can be transmitted to future generations.
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

143. Assertion(A): RuBisCO enzyme is present in bundle sheath cells of C_4 plants performing carboxylation activity where photorespiration does not occur
Reason(R): Photorespiration is suppressed in C_4 plants due to high CO_2 concentration at RuBisCO site and increase photosynthetic rate.
(1) (A) is true but (R) is false.
(2) (A) is false, but (R) is true.
(3) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(4) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

144. Match the contents of List I with List II and choose the correct answer.

	List I		List II
A	Alpha lactalbumin	I	To study Alzheimer's disease
B	Genetically engineered Insulin	II	Reduces the risk of allergic reactions
C	α -interferons	III	Dominant whey protein in human milk
D	Transgenic animals	IV	Cancer treatment

(1) A - II, B-I, C-IV, D-III
(2) A - III, B-I, C-IV, D-II
(3) A-III, B-II, C-IV, D -I
(4) A-I, B-II, C-IV, D-III

145. Which of the following statements is incorrect w.r.t. ETS of respiration
(1) ATP is synthesized through complex V.
(2) Oxidation – reduction reactions produce proton gradient in respiration, between inter membrane space and matrix.
(3) Role of oxygen is limited to the terminal stage in aerobic respiration.
(4) In ETS one molecule of NADH + H^+ gives rise to 2ATP and one $FADH_2$ gives rise to 3ATP.

146. The molecular diagnostic technique used to detect pathogens by the detection of antigens or antibodies is
(1) ELISA test (2) southern blotting
(3) PAP smear test (4) autoradiography

147. The mobile carrier of electrons between complex III and IV of mitochondrial ETS is
(1) PC (2) PQ
(3) *cyt* c (4) *cyt* a_3

148. Without the involvement of natural selection, occasionally the allelic frequencies in a population may be subjected to change due to some chance events. An example for this is
(1) genetic recombination
(2) punctuated equilibrium
(3) founder effect
(4) gene flow

149. Development is considered as the sum of
(1) Growth and senescence
(2) Expansion and cell division
(3) Growth and differentiation
(4) Growth and cell division

150. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A): Ketonuria is one of the symptoms of diabetes mellitus.

Reason (R): Deficiency of ADH causes diabetes mellitus.

In light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true

151. Statement-I: Gibberellins application increases sugarcane yield as much as 20 tonnes per acre.

Statement-II: Gibberellins also promote bolting in beet, cabbage and many plants having rosette habit.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

152. Read the following and choose the correct statements.

I. Natural selection in which more individuals acquire value other than the mean character value is termed directional selection.

II. Jawless fish evolved nearly 350 mya.

III. *Homo habilis* was first to use hides to protect the body.

IV. Adaptive ability is never heritable.

(1) II and III only (2) I and II only

(3) II and IV only (4) I, II and IV

153. Which hormone promotes internode/petiole elongation in deep water rice?

(1) Kinetin (2) Ethylene

(3) 2, 4 – D (4) GA3

154. Given below are two statements.

Statement I : When reptiles came down, birds took over this earth.

Statement II : Small-sized reptiles of the Mesozoic Era still exist today.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct

155. How many single heterozygous daughter plants will be produced in F_2 generation of a dihybrid cross of Mendel?

(1) 8/16 (2) 2/16

(3) 1/16 (4) 4/16

156. Which of the following conditions increases binding of O_2 to hemoglobin at the lung surface?

(1) high pCO_2

(2) low H^+ concentration

(3) low pO_2

(4) high temperature

157. Which of the following statement is *incorrect* regarding linkage?

(1) Linkage is an exception to Mendel's law of independent assortment.

(2) Linkage reduces the chances of recombination.

(3) Closer the linked genes, stronger the tendency of linkage.

(4) The number of linkage groups in an organism is equivalent to number of chromosomes of its diploid somatic cell.

158. When diaphragm contracts, the volume of the thoracic chamber

(1) increases in the vertical axis

(2) increases in the transverse axis

(3) decreases in the antero-posterior axis

(4) decreases in the dorsoventral axis

159. Three genes A, B, C control skin colour in humans. The genotype with all the recessive alleles (aabbcc) will have one of the following skin colour in the individuals.

(1) Light skin colour

(2) Fairly light skin colour

(3) Lightest skin colour

(4) Intermediate skin colour

160. Which of the following electrical events of the heart during cardiac cycle is *not* represented by a distinct wave in a standard ECG?

(1) Atrial depolarisation

(2) Ventricular depolarisation

(3) Atrial repolarisation

(4) Ventricular repolarisation

161. The experiment of whom of the following provided an unequivocal proof that DNA is the hereditary material.

(1) Alfred Hershey and Martha chase.

(2) Avery, Macleod and Mc Carty.

(3) M. Wilkins and Franklin.

(4) Frederick Griffith.

162. Read the following statements regarding the events of cardiac cycle and choose the option with the correct statements.

A. End of atrial systole is the end of ventricular diastole.

B. End of ventricular systole is the beginning of joint diastole.

C. End of joint diastole is the end of atrial diastole.

D. End of joint diastole is the beginning of atrial systole.

(1) A and B only

(2) B and C only

(3) A, B, C and D

(4) A, B and D only

163. Which of the following is *incorrect* statement regarding DNA replication?

(1) Deoxyribonucleoside triphosphates serve a dual purpose. In addition to acting as substrates, they provide energy for polymerization reactions during DNA replication.

(2) DNA dependent DNA polymerase catalyses the polymerization in only one direction, that is 5'→3'.

(3) In eukaryotes, DNA replication takes place in the M phase of the cell cycle.

(4) The discontinuously synthesized fragments are later joined by DNA ligase.

164. *Hemidactylus* is

(1) uricotelic (2) ammonotelic

(3) ureotelic (4) aminotelic

165. In Lac Operon system, which of the following is constitutively synthesized ?

(1) Inducer

(2) β-galactosidase

(3) Transacetylase

(4) Repressor protein

166. The first 7 pairs of ribs in man are

(1) bicephalic and vertebrochondral type

(2) bicephalic and vertebrosternal type

(3) monocephalic and vertebrochondral type

(4) monocephalic and vertebrosternal type

167. Statement-I: In *Escherichia coli* the positively charged DNA is held with some negatively charged proteins in nucleoid region.

Statement-II: Viruses having RNA genome and having shorter life span mutate and evolve faster.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

168. White muscle fibres are characterised by

(1) high myoglobin content

(2) low resistance to fatigue

(3) slow contraction rate

(4) poorly developed sarcoplasmic reticulum

169. Post-transcriptional modification of primary transcript in eukaryotes does not involve.

(1) Removal of exons and joining of introns.

(2) Addition of 7-methyl guanosine triphosphate at 5' end of 'hnRNA'.

(3) Addition of 200-300 adenylate residues at 3' end of 'hnRNA'.

(4) Removal of introns and joining of exons in a coordinated manner.

170. Given below are two statements.

Statement I: The rise in the stimulus-induced influx of Na^+ is extremely short- lived.

Statement II: This permeability to Na^+ is quickly followed by a rise in permeability to K^+.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct

171. Read the following statements with respect to the "Lac Operon", and select the correct option.

A) It has three structural genes.

B) Glucose can act as inducer.

C) Repressor protein is acting in the presence of Lactose.

D) Repressor is synthesized by regulator gene.

(1) Only A and C are correct.

(2) Only B is incorrect.

(3) A and D are correct.

(4) A, B and D are correct.

172. Given below are two statements.

Statement I: LH stimulates androgen secretion in males, and promotes ovulation as well as the formation and maintenance of the corpus luteum in females.

Statement II: FSH and androgens regulate spermatogenesis.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is incorrect, but statement II is correct.

(3) Statement I is correct, but statement II is incorrect.

(4) Both statement I and statement II are incorrect

173. During the process of separation of DNA, chilled ethanol is added to

(1) Precipitate DNA

(2) Break open the cell to release DNA

(3) Remove proteins such as histones

(4) Facilitate action of restriction enzyme

174. Which of the following is a common physiological effect of thyroid hormones, androgens, cortisol and erythropoietin?

(1) Protein catabolism

(2) Anti-inflammatory action

(3) Stimulation of RBC production

(4) Regulation of blood calcium levels

175. Match the following

Column-I		Column-II	
A	PCR	1	Separation and purification of products
B	Gel electrophoresis	2	Amplification of gene of interest
C	Downstream process	3	Separation and isolation of DNA fragments
D	Bioreactors	4	Culturing the host cells

(1) A – 2, B – 3, C – 1, D – 4

(2) A – 1, B – 3, C – 4, D – 2

(3) A – 2, B – 4, C – 3, D – 1

(4) A – 3, B – 4, C – 2, D – 1

176. Under different selection pressures, some organisms produce a large number of small-sized offspring, while others produce fewer but larger offspring. This difference represents strategies to achieve

(1) low r value

(2) high r value

(3) high K value

(4) high mortality value

177. Which one of the following statements is not true regarding "gel electrophoresis" technique?

(1) The presence of chromogenic substance gives blue coloured DNA bands on the gel.

(2) Bright-Orange coloured bands of stained DNA can be observed in the gel when exposed to 'UV light'.

(3) The process of extraction of DNA fragments from gel is called elution.

(4) The DNA fragments are stained by using ethidium bromide.

178. The first letter for naming the restriction enzyme is the

(1) First letter of species of source bacteria.

(2) First letter of genus of the source bacteria.

(3) Strain of bacteria.

(4) First letter of subspecies of the source bacteria.

179. In 'pBR322', the restriction site for "PvuII" is present in

(1) Ampicillin resistance gene.

(2) Tetracycline resistant gene.

(3) Replication of plasmid (rop) gene.

(4) Origin of replication.

180. Which of the following vectors has been used for introducing nematode specific genes in infected tobacco plants?

(1) Retroviruses

(2) Ti plasmid of *Agrobacterium*

(3) Rop gene

(4) *Meloidegyne incognitia*